RAJALAKSHMI ENGINEERING COLLEGE

Akash M 250701034-2025-CSE

A2

Started on	Tuesday, 25 August 2026, 10:44 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 10:48 AM
Time taken	3 mins 29 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int cmp_asc(const void *a, const void *b) {
    return (*(int *)a - *(int *)b);  // ascending order
}

int cmp_desc(const void *a, const void *b) {
    return (*(int *)b - *(int *)a);  // descending order
}

int main() {
    int n;
    scanf("%d", &n);

    int *A = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++)
        scanf("%d", &A[i]);

    int *B = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++)
        scanf("%d", &B[i]);

    qsort(A, n, sizeof(int), cmp_asc);
    qsort(B, n, sizeof(int), cmp_desc);

    long long sum = 0;
    for (int i = 0; i < n; i++)
        sum += (long long)A[i] * B[i];

    printf("%lld\n", sum);

    free(A);
    free(B);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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